Surname: Haque   Given Name: Amaam

Student Number: 400520641   Section:

2024 Midterm EXAMINATION
ELECENG 3TP3
Instructors: M. Elamien and T. Kirubarajan

YOU ARE RESPONSIBLE FOR ENSURING THAT YOUR COPY OF THE PAPER IS COMPLETE. BRING ANY DISCREPANCY TO THE ATTENTION OF YOUR INVIGILATOR.

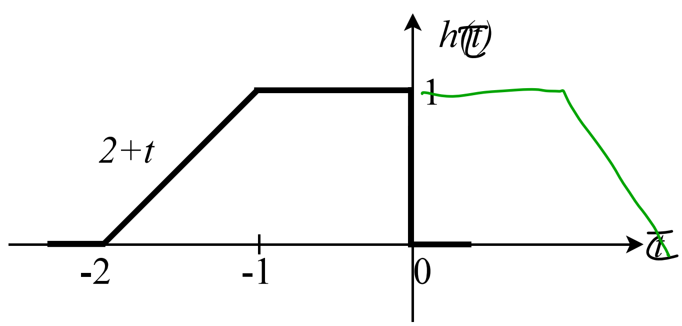
Examination Duration: 50 minutes
Exam Instructions:
SINGLE VERSION exam
Materials Permitted In The Exam Venue:
No electronic aids are permitted, e.g., laptops, phones
McMaster Standard Calculator (Casio FX-991MS/MS+)
One Double Sided `8.5" x 11"` Crib Sheet
Materials To Be Supplied To Students:
None
Instructions To Students:
Answer all questions on the sheets provided.

---

(30) 1. Consider the single pulse $h(t)$ shown below. Define $y(t) = h(t) * h(t)$ as the convolution of $h(t)$ with itself.

$h(t) = 2+t$

$y(t) = h(t) * h(t)$
$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau) \, d\tau$



(a) If $h(t-\tau)$ is the impulse response of a causal linear time invariant system, what values of τ would be permitted?

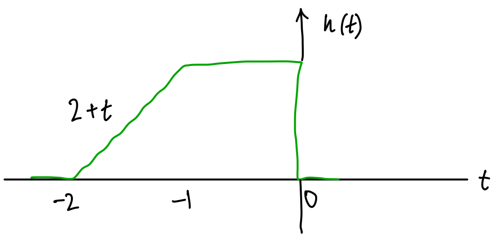
τ limits:

causal: $[0, \infty]$ $\rightarrow$ $\tau = [2, \infty]$

linear :

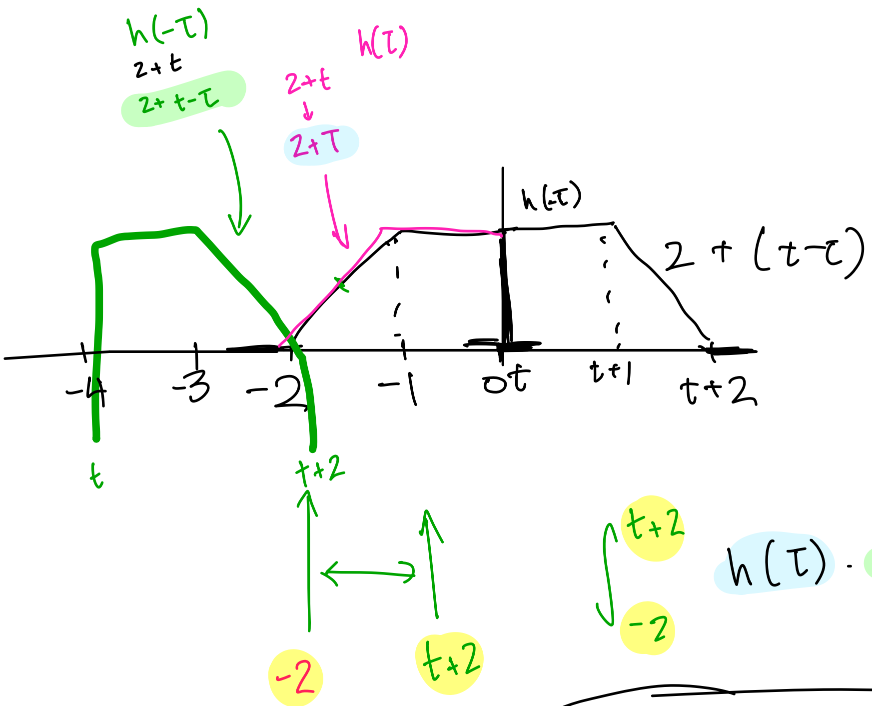
time invarying:

(b) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-4 \le t \le -3$. Do not evaluate the integral.



$$y(t) = h(t) * h(t)$$
$$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau)\, d\tau$$

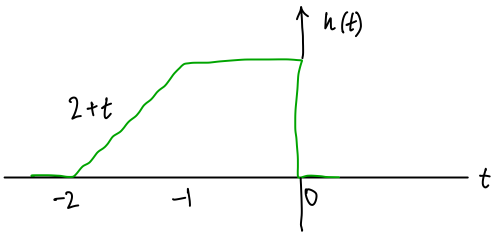
$t=-3$: $\int_{-2}^{t+2} (2+t)$



$$\int_{-2}^{t+2} h(\tau) \cdot h(t-\tau)\, d\tau$$

$$\therefore \int_{-2}^{t+2} (2+\tau) \cdot (2+t-\tau)\, d\tau$$

(b) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-4 \le t \le -3$. Do not evaluate the integral.



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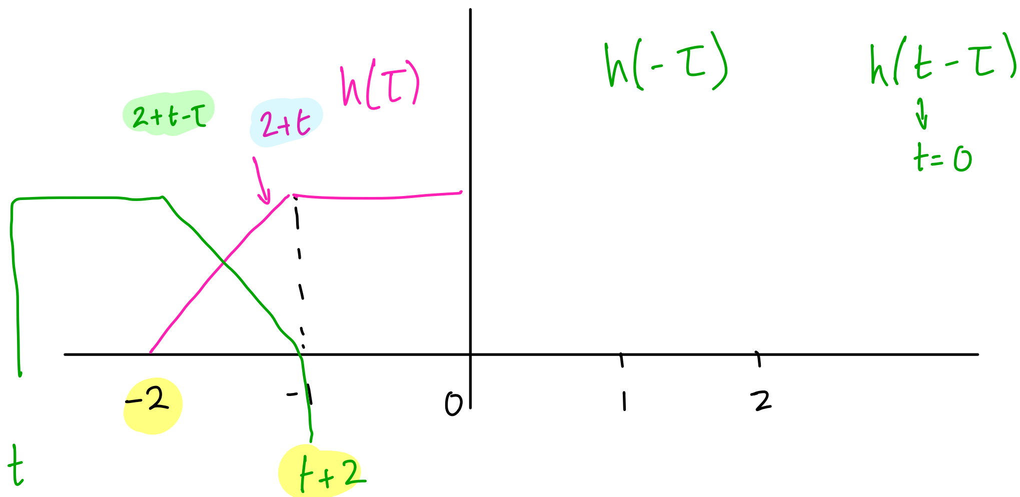
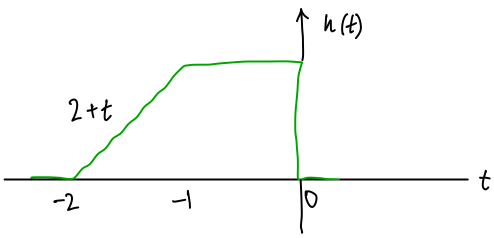


$$\int_{-2}^{t+2} h(\tau) \cdot h(t-\tau)$$

$$\int_{-2}^{t+2} (2+t) \cdot (2+t-\tau)$$

(b) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-4 \le t \le -3$. Do not evaluate the integral.



$y(t) = h(t) * h(t)$

$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau)\, d\tau$

$h(t) = 2 + t$

$h(5) = 2 + 5$

$h(t-\tau) = 2 + t - \tau$

$h(-\tau) \rightarrow h(t-\tau)$

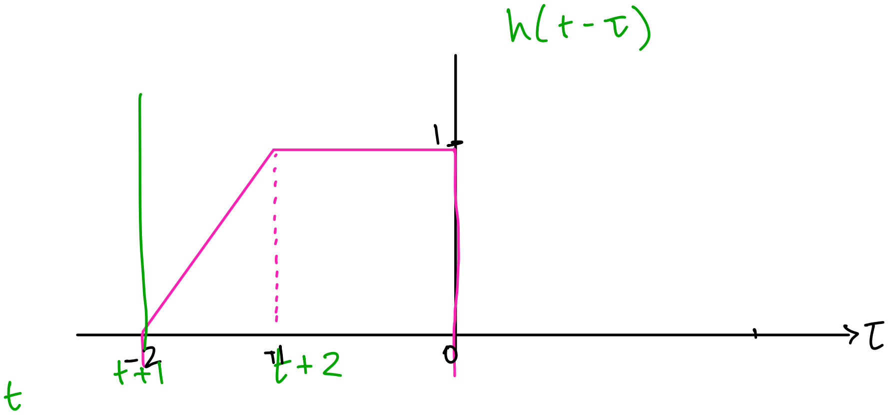
$h(0-\tau)$

$h(t-\tau)$

$\int_{-2}^{t+2} h(\tau) \cdot h(t-2)$

$\therefore \int_{-2}^{t+2} (2+t) \cdot (2+t-\tau)$

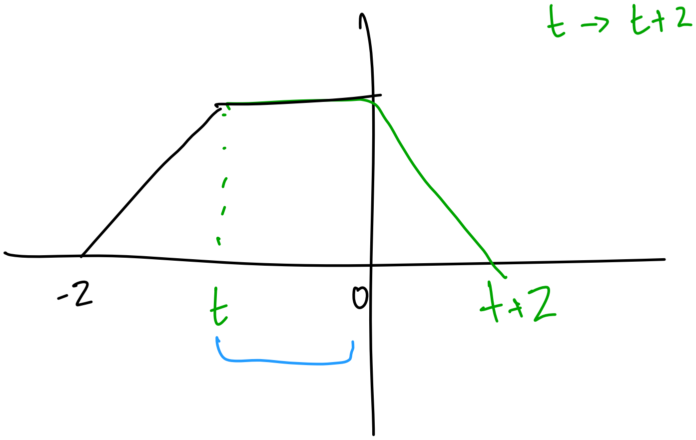
(c) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-3 \le t \le -2$. Do not evaluate the integral.



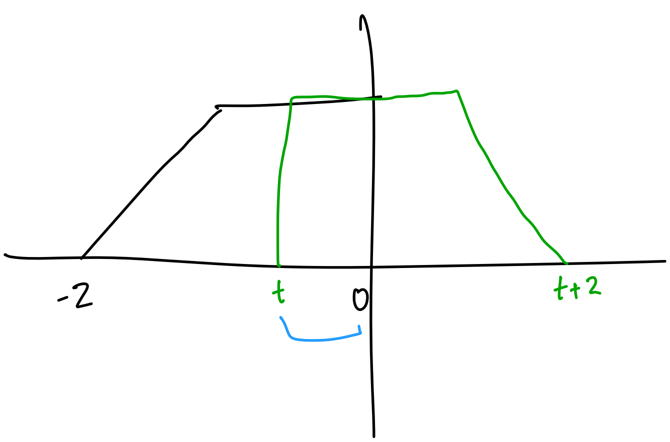
$$c(t) = h(t) * h(t)$$

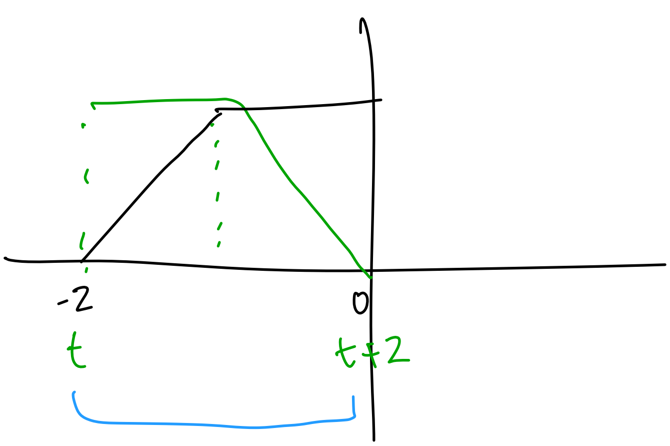
$$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau)$$

$t \to t+2$

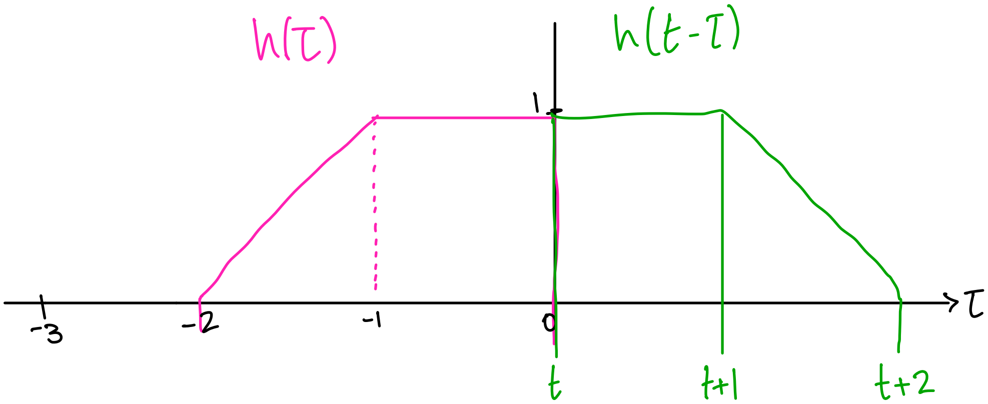


$$= \int_{t}^{0} h(\tau) \cdot h(t-\tau)$$

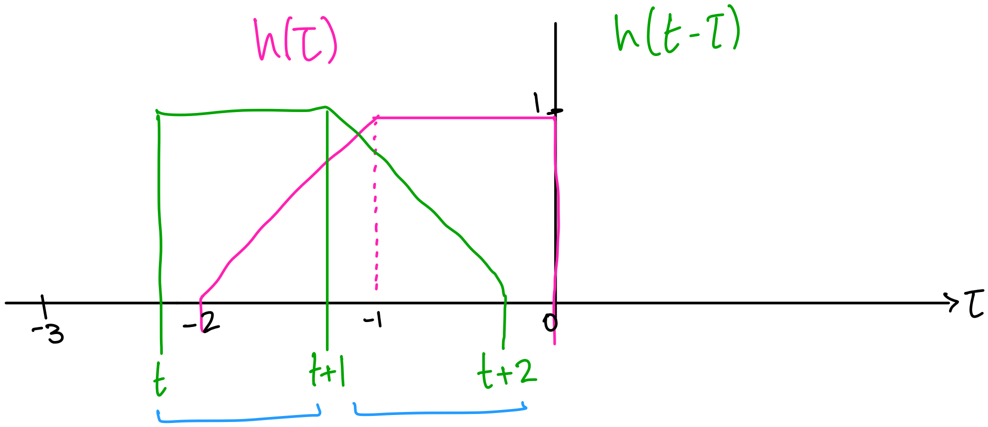




(c) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-3 \leq t \leq -2$. Do <u>not</u> evaluate the integral.



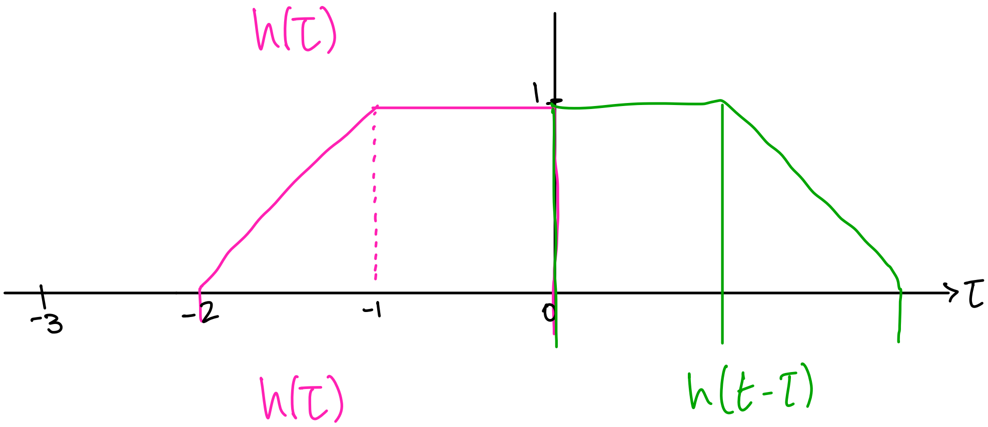
$$c(t) = h(t) * h(t)$$
$$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau)$$

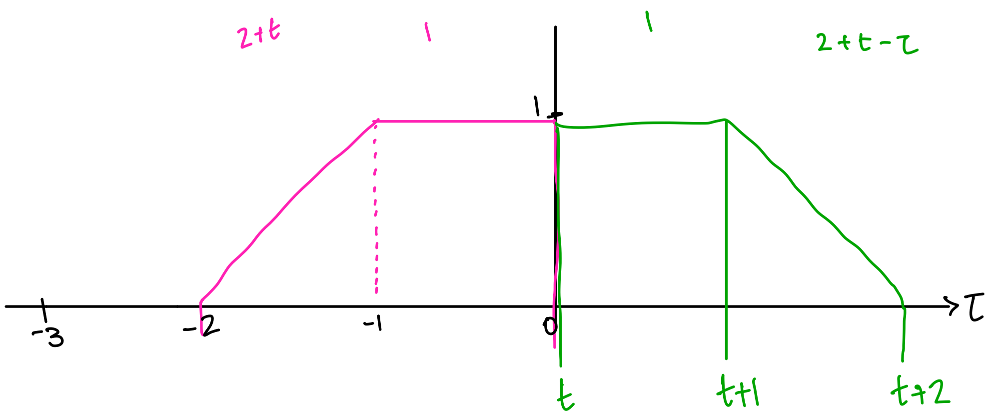


$$\int_{-2}^{t+2} h(\tau)\ h(t-\tau)$$

$$\int_{-2}^{t+1} (2+\tau)(1) + \int_{-1}^{t+2} (1)(2+t-\tau) + \int_{t+1}^{-1} (2+\tau)(2+t-\tau)$$

(c) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-3 \leq t \leq -2$. Do not evaluate the integral.

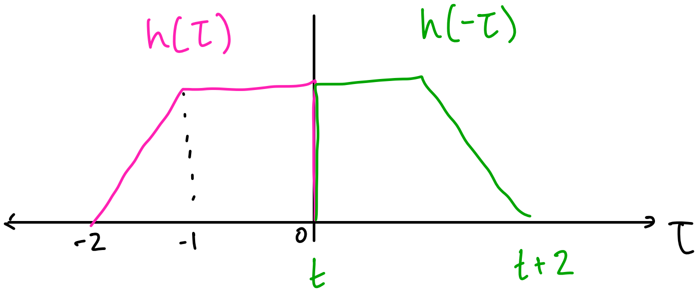




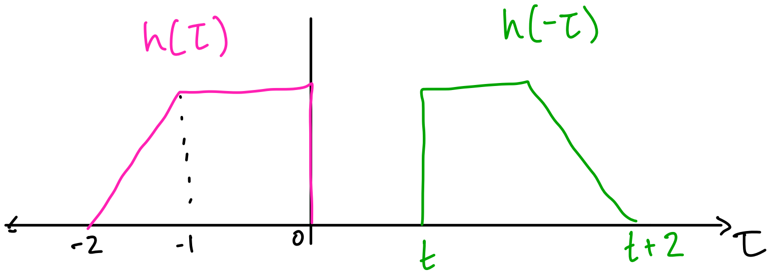
$$\int_{-2}^{t+2}$$

$$= \int_{-2}^{t+1} (2+t)(1) \, d\tau + \int_{-1}^{t+2} (1)(2+t-\tau) \, d\tau + \int_{t+1}^{-1} (2+t)(2+t-\tau) \, d\tau$$

(d) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $1 \leq t \leq 2$. Do <u>not</u> evaluate the integral!



$$y(t) = h(t) * h(t)$$
$$= \int_{-\infty}^{\infty} h(\tau) \cdot h(t-\tau)$$
$$= 0$$



(e) **Bonus Question (5 Marks)**. Evaluate the integrals in parts (b), (c), and (d).

(10) 2. Determine whether the following discrete-time system is causal or noncausal, have memory or is memoryless, is linear or nonlinear, is time invariant or time varying. Justify your answers. In the following system, $x[n]$ is an arbitrary input and $y[n]$ is the response to $x[n]$.

$$y[n] = \frac{1}{4}x[n] + 2x[n+1]$$

i) (non) causal?

non causal.
system depends on input from future $(n+1)$.

ii) (non) linear?

linear. system has additivity & homogeneity.

$a_1 y_1 + a_2 y_2 = a_1 x_1 + a_2 x_2$

• additivity:

↳ if $x_1(t) \to y_1(t)$ → $y_1[n] = \frac{1}{4}x_1[n] + 2x_1[n+1]$

$x_2(t) \to y_2(t)$ → $y_2[n] = \frac{1}{4}x_2[n] + 2x_2[n+1]$

then $x_1(t) + x_2(t) \to y_1(t) + y_2(t)$ → $y_1[n] + y_2[n] = \frac{1}{4}x_1[n] + 2x_1[n+1] + \frac{1}{4}x_2[n] + 2x_2[n+1]$

$= \frac{1}{4}(x_1+x_2)[n] + 2(x_1+x_2)[n+1]$ ✓

• homogenity:

↳ if $x(t) \to y(t)$ then $cx(t) \to cy(t)$ → $cy[n] = c\left(\frac{1}{4}x[n] + 2x[n+1]\right)$

$= \frac{c}{4}x[n] + 2c\,x[n+1]$

iii) memory(less)?

memory. system depends on non current values.

↳ eg. value 'n' depends on 'n-1'.

iv) time (in)variant?

time invariant. if the system has shifted input, then shifted output too.

$y[n-a] = \frac{1}{4}x[n-a] + 2x[n-1-a]$

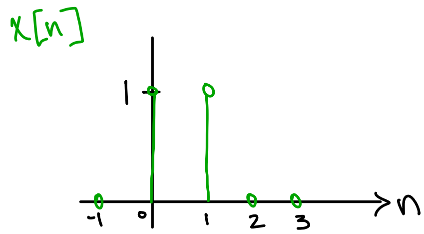
input shifted by a, output shifted by a.

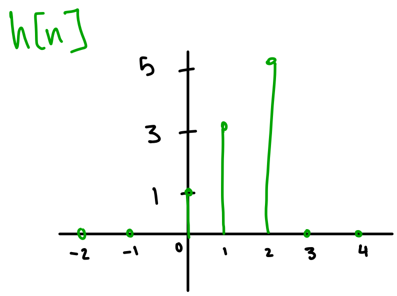
(25) 3. Consider the following discrete-time signals

$$x[n] = u[n] - u[n-2] \longrightarrow [0,2]$$
$$h[n] = \delta[n] + 3\delta[n-1] + 5\delta[n-2] \longrightarrow [0,2]$$

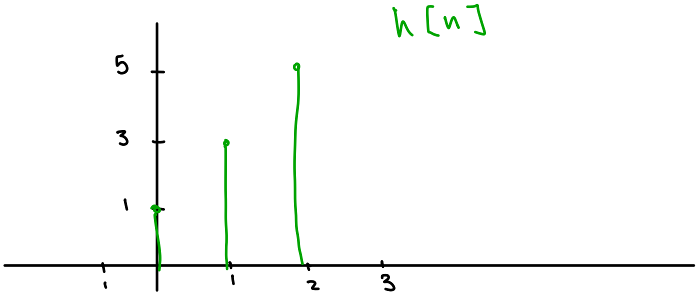
where $u[n]$ and $\delta[n]$ are the discrete-time unit-step function and the unit-pulse function, respectively.

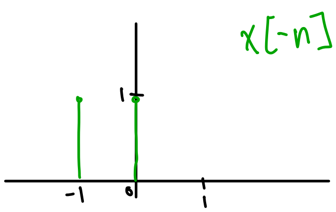
(a) Sketch a stem plot of $x[n]$ and $h[n]$. Make sure you label the values.





(b) Compute and sketch a stem plot of the convolution $y[n] = x[n] * h[n]$. Make sure you label the values.





END OF EXAMINATION

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In Class Stuff

$$x[n] = u[n] - u[n-2] = \delta[n] + \delta[n-1]$$

$$y[n] = x[n] * h[n]$$
$$= h[n] * x[n]$$
$$= h[n] * (\delta[n] + \delta[n-1])$$
$$= h[n] + h[n-1]$$

$h[n] * \delta[n] = h[n]$

Surname: ____________________ Given Name: ____________________

Student Number: ____________________ Section: ____________________

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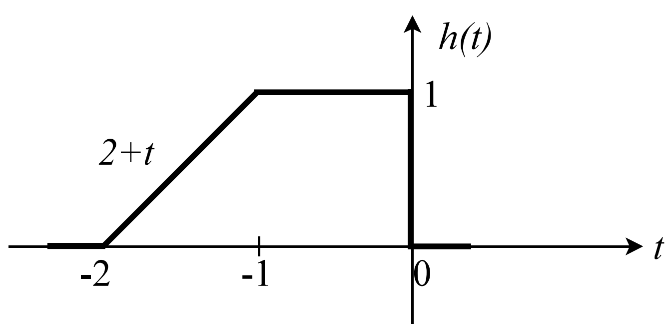

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$$y[n] = x[n] * h[n]$$
$$= h[n] * x[n]$$
$$= h[n] * (\delta[n] + \delta[n-1])$$
$$= h[n] + h[n-1]$$

$h[n] * \delta[n] = h[n]$

(b) Compute and sketch a stem plot of the convolution $y[n] = x[n] * h[n]$. Make sure you label the values.

END OF EXAMINATION